

## **SLIDE 1**

I looked into the effects that artificial intelligence has on human creativity and originality, specifically the ways in which AI is causing a shift in human creativity and how that shift raises concerns for originality and authorship of generated content.

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## **SLIDE 2**

AI is becoming increasingly common in creative work so the big question is whether or not it is pushing the idea of creativity in a forward direction as a tool or if its path might eventually replace it. Through my research I found that rather than eliminating creativity, AI is reshaping it.

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## **SLIDE 3**

Ivcevic and Grandinettie introduce this 4C- Model which separates and defines 4 levels of creativity that are based on performance and application.

Mini-c creativity takes place during learning processes and is subjective to the individual. It's described as having "personal meaningfulness". Examples might include learning how to tie your shoes, where each individual might think about the process in a unique way to accomplish the same result.

Little-c creativity is your everyday challenges where you might not be learning something new but you are tasked with decision making processes. This level is often measured with divergent thinking tests or problem-solving tasks. Many studies involving AI research use little-c creativity as a focal point.

Big-c creativity can sometimes be referred to as eminent creativity and influences larger populations such as a field of study or culture.

And then lastly we have Pro-c creativity or professional creativity which serves as your highest level and requires mastery level content such as original work that can be determined and published. So professional design or artwork would best fit under this category.

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## **SLIDE 4**

In 2023 a study was done by Doshi & Hauser that focused on little-c creativity. Participants were asked to prepare an 8-sentence story about an adventure on an open sea, one set had access to ChatGPT-4 and the other set was not given access to any form of AI. This and many other studies have come up with similar results, and that is that the set of participants who are given access to some form of AI come up with objectively better results. Consensual Assessment Techniques were used to evaluate the results as being original, novel, useful, surprising, or having good overall quality. This Method was developed by Teresa Amabile and is widely used in creativity research. Divergent thinking tests measure individual creativity prior to the study to assess word association and tasks like unusual uses tests. A few categories considered for this specific case were "better written", "more enjoyable", "having a surprising twist", and "having changed people's expectations of future stories they read".

Although studies often show results as having less variability they did find that originality and usefulness was supported at a higher level than those who were seen to have lower creative thinking ability.

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## **SLIDE 5**

So now that we have briefly defined how AI affects creativity, we can now take a look at the issues involving originality. With the involvement of AI, the definition of originality becomes a bit harder to define as the line between human creativity and AI generated work is extremely vague. This idea raises questions as to who actually gets credit for or owns the final product.

A few studies have shown that people value creative work less if they are aware that AI was used in the production process which is interesting to note since this shows that we might naturally agree that AI generated content is not original to the human artist. This is also why this concern increases at higher level professions involving pro-c creativity where originality actually matters.

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## **SLIDE 6**

Although there are those with these concerns, many researchers argue that AI should be viewed as a tool rather than a replacement for creativity, specifically at the pro-c level.

Take other creative tools for example, photoshop or cameras in photography. Once there were similar concerns but are now widely adopted as a form of an artistic medium.

AI can also help to generate ideas, improve efficiency, and even offer new creative directions. This is true for even the lower levels of creativity, such as a teacher making a lesson plan.

So rather than removing creativity, it might just be changing what creativity looks like.

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## **SLIDE 7**

So now the bigger question is whether or not AI is enhancing creativity or simply replacing it

At the lower levels of creativity the results show that it helps people work more efficiently and produce higher quality results, which is a desirable result for professions that aren't concerned with authorship.

It might even help those people who are rated at lower creative levels to produce better work which could expand job opportunities in the job market.

The real downside here is the concern with pro-c professions like design or art where originality is central. An optimal solution is a hybrid approach where human creativity is used with AI as a collaborator.

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## **SLIDE 8**

I'll just touch on this briefly but there are legal and ethical concerns that involve issues with copyright. Who owns the content that is created using AI? Is the user who wrote the prompt or the company who built the AI?

Another important thing to consider is AI training, since the models learn from existing data the generated content is just regurgitated work and then you run into copyright issues there

As of now this topic is largely untapped so there aren't really any legal standards for this. We would need further research and attention moving forward.

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## **SLIDE 9**

As there are many unanswered questions regarding these topics it is very clear that more research needs to be completed in order to answer them.

I suggest implementing a stronger definition of originality, especially in the context of AI-assisted work.

An interesting route to take would be to study the long term effects - does relying on AI affect how people think or create over time?

Another untouched area of research is understanding how training data impacts results, since that has a direct impact on issues of human bias or ownership.

Overall, this is still an underdeveloped field and we are just at the start of understanding its impact.

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## **SLIDE 10**

While AI raises concerns about originality and authorship, there is still no evidence suggesting that it eliminates human creativity. Instead it reshapes how creativity might look in the near future. Based on this I would argue that we should implement a redefinition of creativity in a way that includes AI as a creative tool or medium rather than a replacement. This would include AI in the creative process as something that can expand human potential rather than limit it.